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Wiring substrate

Abstract

A wiring substrate (**1**) includes an insulating substrate (**11**), a surface metal layer provided on a surface of the insulating substrate (**11**), and a wiring layer disposed in an inner part of the insulating substrate (**11**). The surface metal layer (for example, a lead wiring portion (**52**)) is divided by a groove (**61**). The wiring layer (for example, an internal wiring trace (**32**)) is disposed in such a manner that, as viewed from above, the wiring layer detours around a region where the groove (**61**) is formed so as not to overlap with the region where the groove (**61**) is formed.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a wiring substrate in which a groove or an insulating coat layer which can serve as a mark for cutting is formed on a surface metal layer.

BACKGROUND ART

(2) In a wiring substrate on which an electronic component such as a semiconductor element is mounted, a metal layer formed of an electrically conductive material such as metal is provided on a surface of an insulating substrate formed of a ceramic material or the like. This metal layer is

utilized as terminals for connection with external electronic components.
(3) A plating layer for protecting the metal layer is formed on the surface of the metal laver. The plating layer is formed by means of electrolytic plating. Electrolytic plating is performed by supplying electricity to a wiring trace for plating which is electrically connected to the metal layer. In some cases, after formation of the plating layer on the surface of the metal layer in this manner, a portion between the metal layer and the wiring trace for plating must be cut so as to electrically isolate the metal layer from the wiring trace for plating.
(4) For example, Patent Literature 1 describes a technique of grinding and removing a part of a metalized layer by using a grinding apparatus such as Leutor, thereby electrically isolating a metalized wiring layer and a metalized metal layer. Also, Patent Literature 2 describes a technique of cutting a lead wiring trace, serving as a wiring trace for plating, by applying a light beam such as an ion beam, a laser beam, or an electron beam.
CITATION LIST
Patent Literatures
(5) Patent Literature 1: JPH09-022958A Patent Literature 2: JP2017-032290A
SUMMARY OF INVENTION
Technical Problem
(6) In the case where, as described above, a portion between the metal layer and the wiring trace for plating is cut so as to electrically isolate the metal layer from the wiring trace for plating, for example, if the intensity of the light beam applied to the surface of the substrate is excessively high, the light beam may reach an inner part of the substrate, thereby forming a deep groove in the substrate. If a wiring layer or the like formed in the inner part of the substrate overlaps with the region where the groove is formed, a groove is also formed in the wiring layer of the inner part. If the wiring layer is cut in this manner, a conduction failure of the wiring layer occurs. Also, if a groove is formed in the metal layer formed in the inner part of the substrate, a portion of the metal layer is exposed to the surface, so that the metal layer becomes more likely to be shorted.
(7) In view of the above, in one aspect of the present disclosure, an object is to provide a wiring substrate which can reduce the possibility of cutting a metal layer, such as a wiring layer, formed in an inner part of a substrate.
Solution to Problem
(8) A wiring substrate according to one aspect of the present disclosure comprises an insulating substrate, a surface metal layer provided on a surface of the insulating substrate and divided by a groove formed on the insulating substrate, and a wiring layer disposed in an inner part of the insulating substrate. In this wiring substrate, the wiring layer is disposed in such a manner that, as viewed from above, the wiring layer detours around a region where the groove is formed so as not to overlap with the region where the groove is formed.
(9) The above-described configuration can reduce the possibility that the wiring layer formed in an inner part of the insulating substrate is cut in a groove forming step performed for forming the groove which divides the surface metal layer.
(10) In the above-described wiring substrate according to the one aspect of the present disclosure, an insulating coat layer may be provided around the groove.
(11) The above-described configuration enables utilization of the insulating coat layer as a mark for forming the groove.
(12) In the above-described wiring substrate according to the one aspect of the present disclosure, the wiring layer may be disposed in such a manner that, as viewed from above, the wiring layer detours around a region where the insulating coat layer is formed so as not to overlap with the region where the insulating coat layer is formed.
(13) In the above-described configuration, since the wiring layer is provided in such a manner that, as viewed from above, the wiring layer does not overlap with the region where the insulating coat layer serving as a mark for forming the groove is formed, it is possible to further reduce the

possibility that the wiring layer formed in the inner part of the insulating substrate is cut in the groove forming step.
(14) In the above-described wiring substrate according to the one aspect of the present disclosure, an internal metal layer may be further provided in an inner part of the insulating substrate, wherein the internal metal layer has an opening formed in a region which overlaps with the region where the groove is formed as viewed from above in such a manner that the opening contains the entirety of the region where the groove is formed.
(15) The above-described configuration can reduce the possibility that a groove is formed in the internal metal layer, formed in an inner part of the insulating substrate, in the groove forming step performed for forming the groove which divides the surface metal layer. As a result, it is possible to prevent exposure of a portion of the internal metal layer to the surface, thereby reducing the possibility that the internal metal layer is shorted.
(16) In the above-described wiring substrate according to the one aspect of the present disclosure, the wiring layer may be provided at a position located 0.5 mm or less from the surface of the insulating substrate.
(17) By virtue of the above-described configuration, it is possible to dispose the wiring layer, while detouring around the region where the insulating coat layer is formed, at a position located 0.5 mm or less from the surface of the insulating substrate that can be reached by the energy of a laser beam applied to the surface metal layer. Therefore, the possibility that the wiring layer is cut by the laser beam can be avoided.
(18) In the above-described wiring substrate according to the one aspect of the present disclosure, the internal metal layer may be provided at a position located 0.5 mm or less from the surface of the insulating substrate.
(19) By virtue of the above-described configuration, it is possible to avoid disposition of the internal metal layer in a region which overlaps with the region where the insulating coat layer is formed, as viewed from above, at a position located 0.5 rum or less from the surface of the insulating substrate that can be reached by the energy of a laser beam applied to the surface metal layer. Therefore, the possibility that a groove is formed in the internal metal layer can be avoided.
Advantageous Effect of Invention
(20) The wiring substrate according to the one aspect of the present disclosure can reduce the possibility that a metal layer, such as a wiring laver, formed in an inner part of the substrate is cut.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** Schematic plan view showing the structure of a wiring substrate according to one embodiment,
(2) FIG. **2** Schematic plan view showing the structure of a semiconductor package in which a semiconductor element is mounted on the wiring substrate shown in FIG. **1**.
(3) FIG. **3** Plan view showing the configuration of an upper surface of a portion of the wiring substrate shown in FIG. **1**.
(4) FIG. **4** Sectional view showing the structure of a portion of the wiring substrate shown in FIG. **3** along a line A-A.
(5) FIG. **5** Plan view showing a solid pattern provided on a ceramic layer in an inner part of the wiring substrate shown in FIG. **3**.
(6) FIG. **6** Plan view showing internal wiring traces provided on a ceramic layer in an inner part of the wiring substrate shown in FIG. **3**.
(7) FIG. **7** Plan view showing a state in which a groove is formed in an alumina coat forming region of the wiring substrate shown in FIG. **3**.

(8) FIG. **8** Sectional view showing the structure of a portion of the wiring substrate shown in FIG. **7** along a line B-B.
(9) FIG. **9** Sectional view showing the structure of a wiring substrate according to a modification.
(10) FIG. **10** Plan view showing the configuration of an upper surface of a portion of a wiring substrate according to a second embodiment.
(11) FIG. **11** Sectional view showing the structure of a portion of the wiring substrate shown in FIG. **10** along a line E-E.
(12) FIG. **12** Plan view showing a solid pattern and internal wiring traces provided on a ceramic layer in an inner part of the wiring substrate shown in FIG. **10**.

DESCRIPTION OF EMBODIMENTS

(13) Embodiments of the present invention will now be described with reference to the drawings. In the following description, identical components are denoted by the same reference numerals. Their names and functions are the same. Therefore, their detailed descriptions will not be repeated.

First Embodiment

(14) In the present embodiment, a wiring substrate **1** will be described as one example of the wiring substrate according to the present invention. A semiconductor chip **71** is mounted on this wiring substrate **1**, whereby a semiconductor package **70** is constituted.
(15) FIG. **1** schematically shows a planar structure of the wiring substrate **1**. The wiring substrate **1** has an approximately quadrangular shape when viewed from above. The outer shape of the wiring substrate **1** is formed by a ceramic substrate (insulating substrate) **11** including a plurality of ceramic layers. A recess **10** having an approximately quadrangular shape when viewed from above is formed in a central portion of an upper surface of the ceramic substrate **11**. A bottom surface of the recess **10** has an element mounting portion **10***a* on which a semiconductor element such as the semiconductor chip **71** is mounted.
(16) The semiconductor chip **71** is mounted on the element mounting portion **10***a* of the wiring substrate **1** having the above-described structure. As a result, the semiconductor package **70** is obtained. FIG. **2** schematically shows the structure of the semiconductor package **70**.
(17) The wiring substrate **1** is electrically connected to the semiconductor chip **71** through bonding wires **72**. Specifically, respective wiring traces **31** of the wiring substrate **1** are electrically connected to respective connection terminals (not shown) of the semiconductor chip **71** by the bonding wires **72**. As a result, it becomes possible to transmit electrical signals between the wiring traces **31** and the semiconductor chip **71**. The wiring traces **31** are formed on the upper surface of a base ceramic layer **20**.
(18) Also, an upper surface metal layer **51** is formed on an upper surface of a side wall of the ceramic substrate **11** formed to surround the recess **10**. In the present embodiment, the upper surface metal layer **51** is formed on a surface of a second ceramic layer **22** located at the uppermost position. The upper surface metal layer **51** is used as a wiring trace for plating at the time of electrolytic plating. The upper surface metal layer **51** has a frame-shaped conductor trace formed to surround the recess **10**. At the time of electrolytic plating, electric power applied to a terminal portion **53** is transmitted to each metal layer via the upper surface metal layer **51**.
(19) The upper surface metal layer **51** has a lead wiring portion **52**. The lead wiring portion **52** is provided to branch off the upper surface metal layer **51** at a predetermined position. The position of the lead wiring portion **52** can be arbitrarily determined in accordance with the shapes of various conductor traces (wirings, terminals, vias, etc.) formed on the wiring substrate **1**.
(20) A terminal portion **53** is provided at a distal end of the lead wiring portion **52**. This terminal portion **53** is used as a connection terminal to which electric power is applied at the time of electrolytic plating. Also, this terminal portion **53** is used as a terminal for inspection at the time of electrical inspection of the wiring substrate **1**.
(21) The surfaces of the upper surface metal layer **51** (including the lead wiring portion **52**) and the terminal portion **53** are covered with a plating layer. The plating layer is formed by means of

electrolytic plating. Electrolytic plating is performed by applying electric power to metal layers such as the upper surface metal layer **51** through the terminal portion **53** and the lead wiring portion **52**. After formation of the plating layer on the surfaces of the metal layers in this manner, the lead wiring portion **52** is cut, whereby the terminal portion **53** and the upper surface metal layer **51** are electrically isolated from each other.
(22) Also, an alumina coat (insulating coat layer) **55** is provided on the lead wiring portion **52** such that the alumina coat **55** extends across a portion of the lead wiring portion **52**. The alumina coat **55** serves as a mark indicating a cutting location at the time of a process of cutting (laser-cutting) the above-described, wiring, which is performed after completion of the plating. The alumina coat **55** is formed by applying an alumina-containing insulating paste at a predetermined position on the lead wiring portion **52**.
(23) Notably, in the present embodiment, the alumina coat is mentioned as one example of the insulating coat layer. However, the material of the insulating coat layer is not limited to alumina and may be any non-conductive material.
(24) In a state in which the semiconductor chip **71** is mounted on the wiring substrate **1**, a liquid resin material is poured into the recess **10** of the ceramic substrate **11**. As a result of hardening of this resin material, a state in which the recess **10** is filled with the resin is created. As a result, the semiconductor package **70** is obtained. Notably, the semiconductor package **70** may have a structure in which no resin material is poured into the recess **10** of the ceramic substrate **11**. Namely, in a different embodiment, the semiconductor package **70** is formed by mounting the semiconductor chip **71** on the wiring substrate **1** and electrically connecting the semiconductor chip **71** and the wiring substrate **1** to each other.
(25) The ceramic substrate **11** has a layered structure in which a plurality of ceramic layers are stacked. In the present embodiment, the ceramic substrate **11** has a structure in which a base ceramic layer **20**, a first ceramic layer **21**, and a second ceramic layer **22** are stacked in this order from the lower side (see FIG. **4**, etc.).
(26) Each ceramic layer can be formed of, for example, a high-temperature fired ceramic material whose main component is alumina (Al.sub.2O.sub.3). In a different embodiment, the ceramic sheet may be formed of meddle-temperature fired ceramic (MICC), such as glass-ceramic, or low-temperature fired ceramic (LTCC).
(27) The base ceramic layer **20** has the shape of an approximately quadrangular flat plate. The recess **10** is formed in a central portion of an upper surface of the base ceramic layer **20**. A bottom surface of the recess **10** serves as the element mounting portion **10***a*.
(28) The first ceramic layer **21** has the shape of an approximately quadrangular flat plate whose size is approximately the same as the base ceramic layer **20**. The first ceramic layer **21** has an opening at its central portion; i.e., has a frame-like shape. The first ceramic layer **21** is stacked on the base ceramic layer **20** such that the first ceramic layer **21** surrounds an outer circumferential portion of the upper surface of the base ceramic layer **20** when viewed from above.
(29) The second ceramic layer **22** has the shape of an approximately quadrangular flat plate whose size is approximately the same as the base ceramic layer **20** and the first ceramic layer **21**. The second ceramic layer **22** has an opening at its central portion; i.e., has a frame-like shape. The opening of the second ceramic layer **22** has an opening area approximately the same as the opening of the first ceramic layer **21**. The second ceramic layer **22** is stacked on the first ceramic layer **21** such that the second ceramic layer **22** surrounds an outer circumferential portion of the upper surface of the first ceramic layer **21** when viewed from above.
(30) By virtue of the above-described structure, in the wiring substrate **1**, the side wall of the ceramic substrate **11** formed by the first ceramic layer **21** and the second ceramic layer **22** is formed to surround the outer circumference of the recess **10** located at the central portion of the upper surface of the base ceramic layer **20**.
(31) Notably, in the present embodiment, each of the first ceramic layer **21** and the second ceramic

layer **22** has an opening at its central portion; i.e., has a frame-like shape. However, in a different embodiment, a structure in which none of the first ceramic layer **21** and the second ceramic layer **22** has no opening is possible.

(32) The wiring traces **31**, the upper surface metal layer **51**, etc. formed on the respective ceramic layers are formed by conductive traces obtained by shaping an electrically conductive material into a predetermined shape. The conductive traces can be formed of a metal material such as copper (Cu), titanium (Ti), tungsten (W), silver (Ag), palladium (Pd), gold (Au), platinum (Pt), molybdenum (Mo), nickel (Ni), or manganese (Mn), or formed of an alloy material which contains any of these metal materials as a main component.

(33) A conventionally known method is used for formation of the conductive traces. Examples of the conventionally known method include a metallization method using printing paste and a method in which a patterned metal layer is transferred. Of these methods, for example, the metallization method is preferably used.

(34) Subsequently, the structure of a portion of the wiring substrate **1** around a region where the lead wiring portion **52** and the terminal portion **53** are formed will be described more specifically. FIG. **3** shows the structure of a portion of the wiring substrate **1** around the lead wiring portion **52**. FIG. **3** is a plan view which shows, on an enlarged scale, a region within a broken line frame of the wiring substrate **1** shown in FIG. **1**. FIG. **4** is a sectional view showing the structure of a portion of the wiring substrate **1** shown in FIG. **3** along a line A-A.

(35) Also, FIG. **5** is a plan view showing the pattern shape of a solid pattern (internal metal layer) **41** provided on the first ceramic layer **21** of the wiring substrate **1** shown in FIG. **3**. FIG. **6** is a plan view showing the pattern shapes of internal wiring traces (wiring layer) **32** provided on the base ceramic layer **20** of the wiring substrate **1** shown in FIG. **3**.

(36) As shown in FIG. **4**, etc., in the present embodiment, the ceramic substrate **11** has a structure in which the base ceramic layer **20**, the first ceramic layer **21**, and the second ceramic layer **22** are stacked in this order from the lower side.

(37) Metal layers such as the terminal portion **53** and the lead wiring portion **52** are formed on a surface of the second ceramic layer **22** which forms the upper surface of the ceramic substrate **11**. These metal layers are also called surface metal layers. Also, the alumina coat **55** is disposed on the lead wiring portion **52** so as to cover a portion of the lead wiring portion **52**. In the present embodiment, the alumina coat **55** is disposed to intersect with the extension direction of the lead wiring portion **52**; namely, to extend across the lead wiring portion **52**.

(38) A lower surface metal layer **35** is formed on a surface of the base ceramic layer **20**, which forms the lower surface of the ceramic substrate **11**. Like the wiring traces **31** and the upper surface metal layer **51**, the lower surface metal layer **35** is formed of a conductor trace Obtained by shaping an electrically conductive material into a predetermined shape. As shown in FIG. **4**, etc., the lower surface metal layer **35** is electrically connected to the terminal portion **53** provided on the upper surface of the ceramic substrate **11** through a via **36** provided to penetrate the ceramic substrate **11**.

(39) Also, conductor traces, such as the internal wiring traces **32** and the solid pattern **41**, are provided in an inner part of the ceramic substrate **11**. Like the wiring traces **31**, the upper surface metal layer **51**, etc., each of the internal wiring traces **32** and the solid pattern **41** is formed of a conductor trace obtained by shaping an electrically conductive material into a predetermined shape.

(40) In the present embodiment, the solid pattern **41** is provided on the first ceramic layer **21**. The solid pattern **41** is provided to cover the greater part of the surface of the first ceramic layer **21**. The solid pattern **41** is connected to, for example, the ground (GND). This can reduce the influence of noise on electric signals transmitted through wiring traces provided in the wiring substrate **1**. Notably, it is preferred that, in order to enhance the noise reduction effect, the area in which the solid pattern **41** is disposed be expanded to an extent possible.

(41) In view of the above, in the wiring substrate **1** according to the present embodiment, the solid pattern **41** is formed over the entire surface of the first ceramic layer **21**, excluding a pattern

removed region (opening) **42**, a region where the via **36** is disposed, and a region therearound.

(42) The pattern removed region **42** is formed by providing an opening in a portion of the conductor trace which forms the solid pattern **41**. The pattern removed region **42** is provided in a region which overlaps with the region where the alumina coat **55** is formed, as viewed from above, such that the pattern removed region **42** contains the entire region where the alumina coat **55** is formed (see FIG. **5**). In the example shown in FIG. **5**, the pattern removed region (opening) **42** has an approximately quadrangular shape. However, the shape of the pattern removed region (opening) **42** is not limited to the approximately quadrangular shape. Also, in the case where the pattern removed region **42** is provided in an end portion of the solid pattern **41**, the pattern removed region **42** may be formed by removing a portion of the end portion of the solid pattern **41**.

(43) Since the pattern removed region **42** as described above is provided in the solid pattern **41**, it is possible to avoid the possibility that, when the lead wiring portion **52** is cut by using, for example, a laser beam, the laser beam reaches the position of the solid pattern **41** and forms a groove in the solid pattern **41**. As a result, even in the case where a groove **61** is formed in the first ceramic layer **21**, exposure of a portion of the solid pattern **41** to the surface can be avoided, whereby the possibility that the solid pattern **41** is shorted can be reduced.

(44) Also, in the present embodiment, the internal wiring traces **32** are provided on the base ceramic layer **20**. The internal wiring traces **32** are routed between the base ceramic layer **20** and the first ceramic layer **21** to have respective arbitrary pattern shapes. Some internal wiring traces **32** are connected to the wiring traces **31** exposed to the upper surface of the wiring substrate **1**.

(45) The internal wiring traces **32** are disposed in such a manner that, as viewed from above, the internal wiring traces **32** detour around the region where the alumina coat **55** is formed so as not to overlap with the region where the alumina coat **55** is formed (see FIG. **6**). As a result, it is possible to avoid the possibility that, when the lead wiring portion **52** is cut by using, for example, a laser beam, the laser beam reaches the position of the internal wiring traces **32**, whereby the internal wiring traces **32** are cut by the laser beam.

(46) Subsequently, the wiring substrate **1** with the groove **61** formed in the lead wiring portion **52** will be described, FIG. **7** shows a state in which the groove **61** is formed in a region of the upper surface of the wiring substrate **1** where the alumina coat **55** is formed. FIG. **7** corresponds to an enlarged view of a region of the semiconductor package **70** within a broken-line frame shown in FIG. **2**. FIG. **8** is a sectional view showing the structure of a portion of the wiring substrate **1** shown in FIG. **7** along a line B-B.

(47) The groove **61** is provided to extend across the lead wiring portion **52**, Such a groove **61** can be formed, after manufacture of the wiring substrate **1** shown in FIG. **1**, by performing a groove forming step in which a light beam (for example, laser beam, ion beam, electron beam, or the like) or a cutting tool (for example, Leutor or the like) is used. The groove forming step is performed after plating layers have been formed, by means of electrolytic plating, on the metal layers formed on the substrate surface, etc. after firing of the wiring substrate **1**. As a result of formation of the groove **61** which divides the lead wiring portion **52** as described above, the electrical connection between the terminal portion **53** and the upper surface metal layer **51** can be cut off.

(48) The groove forming step is preferably performed by using a laser beam. This makes it possible to form a groove while more precisely designating an intended position and to form a finer groove. The laser beam used in the present embodiment is, for example, an ordinary laser beam used in laser machining, such as CO.sub.2 laser, fiber laser, and YAG laser. Also, ordinary conditions employed for cutting wiring traces formed on a substrate can be applied to the conditions of laser machining.

(49) In FIG. **3**, a broken-line frame shows a region C to which a laser beam is applied in the groove forming step. As described above, the alumina coat **55**, which can serve as a mark indicating a cutting location is provided in a to-be-cut region on the lead wiring portion **52**. The region C to which the laser beam is applied is located within the region where the alumina coat **55** is formed.

At the time of the groove forming step, the laser beam is applied to the region where the alumina coat **55** is formed.

(50) When the laser beam is applied to the region C, the groove **61** is formed. As shown in FIG. **8**, etc., the groove **61** is formed to reach an inner part of the ceramic substrate **11**. Although the depth of the formed groove **61** changes with various conditions, such as the intensity of the laser beam, the formed groove **61** may have a depth of, for example, up to about 0.5 mm as measured from the substrate surface.

(51) Meanwhile, the thickness of each of the ceramic layers such as the first ceramic layer **21** and the second ceramic layer **22** falls within the range of, for example, 0.1 mm to 0.3 mm. Therefore, when a laser beam is applied form the upper surface side of the wiring substrate **1**, the laser beam may reach an inner part of the first ceramic layer **21**. As a result, as shown in FIG. **8**, the groove **61** may be formed through the second ceramic layer **22** and formed in a portion of the first ceramic layer **21**.

(52) In the wiring substrate **1** according to the present embodiment, the pattern removed region **42** is provided in the solid pattern **41** provided between the first ceramic layer **21** and the second ceramic layer **22**. As described above, the pattern removed region **42** is provided in a region which overlaps with the region where the alumina coat **55** is formed, as viewed from above, such that the pattern removed region **42** contains the entire region where the alumina coat **55** is formed (see FIG. **5**). Furthermore, the region C to which the laser beam is applied is located within the region where the alumina coat **55** is formed.

(53) Therefore, the groove **61** formed as a result of application of the laser beam is located in the region where the pattern removed region **42** is formed as viewed from above. Namely, the solid pattern **41** has the pattern removed region **42** formed in a region which overlaps with the region where the groove **61** is formed, as viewed from above, in such a manner that the pattern removed region **42** contains the entirety of the region where the groove **61** is formed.

(54) This configuration enables avoidance of the possibility that, at the time of the groove forming step, the laser beam reaches the position of the solid pattern **41** and a groove is formed in the solid pattern **41**. As a result, even in the case where the groove **61** is formed in the first ceramic layer **21**, exposure of a portion of the solid pattern **41** to the surface can be avoided, whereby the possibility that the solid pattern **41** is shorted can be reduced.

(55) Also, in the wiring substrate **1** of the present embodiment, the internal wiring traces **32** are provided between the base ceramic layer **20** and the first ceramic layer **21**. As described above, the internal wiring traces **32** are disposed in such a manner that, as viewed from above, the internal wiring traces **32** detour around the region where the alumina coat **55** is formed, so as not to overlap with the region where the alumina coat **55** is formed (see FIG. **6**). Furthermore, the region C to which the laser beam is applied is located within the region where the alumina coat **55** is formed.

(56) Therefore, the groove **61** formed as a result of application of the laser beam is formed in a region which does not overlap with the positions of the internal wiring traces **32** as viewed from above. Namely, the internal wiring traces **32** are disposed in such a manner that, as viewed from above, the internal wiring traces **32** detour around the region where the groove **61** is formed, so as not to overlap with the region where the groove **61** is formed.

(57) By virtue of this configuration, even when the laser beam reaches the first ceramic layer **21**, on which the internal wiring traces **32** are formed, in the groove forming step, it is possible to avoidance the possibility that the internal wiring traces **32** are cut by the laser beam.

(58) As described above, the groove **61** formed in the groove forming step has a depth of, for example, up to about 0.5 mm as measured from the substrate surface. Therefore, the present embodiment is preferably applied to a structure in which the internal wiring traces **32** are provided at a position located 0.5 mm or less from the surface of the ceramic substrate **11**. Similarly, the present embodiment is preferably applied to a structure in which the solid pattern **41** is provided at a position located 0.5 mm or less from the surface of the ceramic substrate **11**.

(59) Also, the structure of the present embodiment is more preferably applied to a structure in which the internal wiring traces **32** and the solid pattern **41** are provided at respective positions located 0.3 mm or less from the surface of the ceramic substrate **11**. The thickness of each ceramic layer falls within the range of 0.1 mm to 0.3 mm. Accordingly, as to a wiring layer formed on a ceramic layer which is located away from the substrate surface by a distance corresponding to three or more layers, the wiring layer is not required to detour around the region where the alumina coat **55** and the groove **61** are formed. Also, as to an internal metal layer formed on a ceramic layer which is located away from the substrate surface by a distance corresponding to three or more layers, the internal metal layer is not required to have an opening in a region which overlaps with the region where the alumina coat **55** and the groove **61** are formed, as viewed from above.

Modification 1

(60) FIG. **9** shows the sectional structure of a portion of a wiring substrate **101** according to one modification of the present embodiment. The wiring substrate **101** includes a ceramic substrate **111**. The ceramic substrate **111** has a structure in which a base ceramic layer **20**, a third ceramic layer **23**, a first ceramic layer **21**, and a second ceramic layer **22** are stacked in this order from the lower side. The wiring substrate **101** differs from the wiring substrate **1** in the point that the third ceramic layer **23** is further provided between the base ceramic layer and the first ceramic layer **21**.

(61) Conductor traces, such as the internal wiring traces **32** and the solid pattern **41**, are provided in an inner part of the ceramic substrate **111**. As to the configurations of the internal wiring traces **32** and the solid pattern **41**, configurations similar to those employed in the wiring substrate **1** can be applied.

(62) In the wiring substrate **101** according to the modification, a metal layer **135** is further provided between the base ceramic layer **20** and the third ceramic layer **23**. The metal layer **135** may function as a solid pattern or an internal wiring. The distance D from the surface of the ceramic substrate **111** to the position of the metal layer **135** is greater than. 0.5 mm. As viewed from above, the metal layer **135** is provided at a position which overlaps with the region where the groove **61** is formed or the region where the alumina coat **55** is formed.

(63) As described above, at a position where the distance D between that position and the surface of the ceramic substrate **111** in the thickness direction is greater than 0.5 mm, an internal metal layer or a wiring layer such as the metal layer **135** may be provided, because the possibility that a laser beam reaches such a position in the groove forming step is low.

(64) Notably, it is more preferred that, as in the wiring substrate **1** of the first embodiment, on all the ceramic layers of the ceramic substrate **11**, the internal wiring traces **32** be disposed in such a manner that, as viewed from above, they detour around the region where the alumina coat **55** or the groove **61** is formed so as not to overlap with the region where the alumina coat **55** or the groove **61** is formed. Similarly, it is preferred that, on all the ceramic layers of the ceramic substrate **11**, the solid pattern **41** have a pattern removed region **42** in a region which overlaps with the region where the alumina coat **55** or the groove **61** is formed, as viewed from above. By virtue these configurations, it is possible to omit electrical inspection of wirings performed after manufacture of the wiring substrate **1**.

Other Modifications

(65) In different modifications, the alumina coat **55** may be omitted. Namely, in a wiring substrate according to one modification, the groove **61** is provided to extend across the lead wiring portion **52** provided on the surface of the ceramic substrate **11**. However, the alumina coat **55** is not provided around the groove **61**. The internal wiring traces **32** disposed in an inner part of the ceramic substrate **11** are disposed in such a manner that, as viewed from above, they detour around the region where the groove **61** is formed so as not to overlap with the region where the groove **61** is formed.

(66) Also, the terminal portion **53** provided at the distal end of the lead wiring portion **52** is not limited to the connection terminal to which electric power is applied at the time of electrolytic

plating. In one modification, the terminal portion **53** may be a particular mounting pad, for example, a pad for mounting a capacitor. Namely, the process of cutting the lead wiring portion **52** may be performed so as to isolate the particular mounting pad from other mounting pads.

Summary of First Embodiment

(67) As described above, the wiring substrate **1** according to the present embodiment includes the ceramic substrate (insulating substrate) **11**, a surface metal layer provided on the surface of the ceramic substrate **11**, a wiring layer disposed in an inner part of the ceramic substrate **11**, and the groove **61** provided to extend across the surface metal layer. The surface metal layer (for example, the lead wiring portion **52**) is divided by the groove **61**. The wiring layer (for example, the internal wiring traces **32**) is disposed in such a manner that, as viewed from above, the wiring layer detours around the region where the groove **61** is formed so as not to overlap with the region where the groove **61** is formed.

(68) By virtue of the above-described configuration, it is possible to reduce the possibility that the wiring layer, etc. formed in an inner part of the ceramic substrate **11** are cut in the groove forming step performed for forming the groove **61** in the lead wiring portion **52**.

Still Other Modifications

(69) Notably, the wiring substrate **1** is not required to have the groove **61**, for example, in the case where the wiring substrate **1** is shipped as an individual substrate.

(70) Namely, a wiring substrate **1** according to another different modification includes the ceramic substrate (insulating substrate) **11**, a surface metal layer provided on the surface of the ceramic substrate **11**, a wiring layer disposed in an inner part of the ceramic substrate **11**, and an insulating coat layer (for example, the alumina coat **55**) disposed to cover a portion of the surface metal layer (for example, the lead wiring portion **52**). The wiring layer (for example, the internal wiring traces **32**) is disposed in such a manner that, as viewed from above, the wiring layer detours around the region where the insulating coat layer is formed so as not to overlap with the region where the insulating coat layer is formed.

(71) By virtue of the above-described configuration, it is possible to reduce the possibility that the wiring layer, etc. formed in an inner part of the ceramic substrate **11** are cut when a laser beam or the like is applied to the alumina coat **55** on the lead wiring portion **52**, thereby cutting the lead wiring portion **52**.

(72) As described above, the wiring substrate according to the different modification includes an insulating substrate, a surface metal layer provided on the surface of the insulating substrate, a wiring layer disposed in an inner part of the insulating substrate, and an insulating coat layer disposed to cover a portion of the surface metal layer. In this wiring substrate, the wiring layer is disposed in such a manner that, as viewed from above, the wiring layer detours around the region where the insulating coat layer is formed so as not to overlap with the region where the insulating coat layer is formed.

(73) By virtue of the above-described configuration, it is possible to avoid the possibility that, when the surface metal layer is divided by using a laser beam or the like, the laser beam reaches an internal position where the wiring layer is formed, whereby the wiring layer is cut by the laser beam.

(74) In the above-described wiring substrate according to the different modification, an internal metal layer may be further provided in an inner part of the insulating substrate, and the internal metal layer may have an opening provided in a region overlapping with the region where the insulating coat layer is formed, as viewed from above, in such a manner that the opening contains the entire region where the insulating coat layer is formed.

(75) By virtue of the above-described configuration, it is possible to avoid the possibility that, when the surface metal layer is divided by using a laser beam or the like, the laser beam reaches a position where the internal metal layer is formed, whereby a groove is formed in the internal metal layer. As a result, it is possible to prevent exposure of a portion of the internal metal layer to the

surface, thereby reducing the possibility that the internal metal layer is shorted.

Second Embodiment

(76) Subsequently, a wiring substrate **1** according to a second embodiment will be described with reference to FIGS. **10** to **12**. FIG. **10** shows, on an enlarged scale, an upper surface of a portion of the wiring substrate **201** according to the second embodiment. FIG. **10** shows the structure of a portion of the wiring substrate **201** around lead wiring portions **52**.

(77) In the wiring substrate **201** according to the second embodiment, two lead wiring portions **52**, which branch off the upper surface metal layer **51** at predetermined positions are provided on the upper surface of the ceramic substrate **11**. In FIG. **12**, of the two lead wiring portions **52**, the lead wiring portion **52** located on the left side is referred to as the lead wiring portion **52**A, and the lead wiring portion **52** located on the right side is referred to as the lead wiring portion **52**B. A terminal portion **53** is provided at the distal end of each lead wiring portion **52**. In FIG. **12**, of the two terminal portions **53**, the terminal portion located on the left side is referred to as the terminal portion **53**A, and the terminal portion located on the right side is referred to as the terminal portion **53**B.

(78) An alumina coat (insulating coat layer) **55**A is provided on the lead wiring portion **52**A in such a manner that the alumina coat **55**A extends across a portion of the lead wiring portion **52**A. An alumina coat (insulating coat layer) **55**B is provided on the lead wiring portion **52**B in such a manner that the alumina coat **55**B extends across a portion of the lead wiring portion **52**B. The alumina coats **55**A and **55**B serve as marks indicating cutting locations at the time of a process of cutting (laser-cutting) the above-described wiring, which is performed after completion of plating.

(79) A groove **61**A is provided in the region where the lead wiring portion **52**A is formed. Also, a groove **61**B is provided in the region where the lead wiring portion **52**B is formed. The grooves **61**A and **61**B are provided in such a manner that they extend across the lead wiring portions **52**A and **52**B, respectively.

(80) FIG. **11** is a sectional view showing the structure of a portion of the wiring substrate **1** shown in FIG. **10** along a line E-E. FIG. **12** is a plan view showing the pattern shapes of a solid pattern (internal metal layer) **241** and internal wiring traces (wiring layer) **232** provided on the first ceramic layer **21** of the wiring substrate **201** shown in FIG. **10**.

(81) Conductor traces, such as the internal wiring traces **232**, the solid pattern **241**, and the internal wiring traces **32**, are provided in an inner part of the ceramic substrate **11**. Specifically, the internal wiring traces **232** and the solid pattern **241** are provided on the first ceramic layer **21**, and the internal wiring traces **32** are provided on the base ceramic layer **20**.

(82) The solid pattern **241** is provided to cover a partial region of the surface of the first ceramic layer **21**. In the example shown in FIG. **12**, the solid pattern **241** is provided in a region which overlaps with the region where the lead wiring portion **52**A, the terminal portion **53**A, etc. are formed, as viewed from above.

(83) The solid pattern **241** has a pattern removed region **42**. The pattern removed region **42** is formed by providing an opening in a portion of the conductor trace which forms the solid pattern **41**. The pattern removed region **42** is provided in a region which overlaps with the region where the alumina coat **55**A provided on the lead wiring portion **52**A is formed, as viewed from above, in such a manner that the pattern removed region **42** contains the entire region where the alumina coat **55** is formed (see FIG. **12**). A region C to which a laser beam is applied at the time of the groove forming step is located within the region where the alumina coat **55**A is formed.

(84) This configuration enables avoidance of the possibility that, at the time of the groove forming step, the laser beam reaches the position of the solid pattern **41** and a groove is formed in the solid pattern **41**. As a result, even in the case where the groove **61**A is formed in the first ceramic layer **21**, exposure of a portion of the solid pattern **241** to the surface can be avoided, whereby the possibility that the solid pattern **241** is shorted can be reduced.

(85) Also, in the present embodiment, not only the solid pattern **241** but also the internal wiring

traces **232** are provided on the first ceramic layer **21**. The internal wiring traces **232** are disposed in such a manner that, as viewed from above, the internal wiring traces **232** detour around the region where the alumina coat **55**B is formed so as not to overlap with the region where the alumina coat **55**B is formed (see FIG. **12**), Another region C to which a laser beam is applied at the time of the groove forming step is located within the region where the alumina coat **55**B is formed.
(86) By virtue of this configuration, the groove **61**B formed as a result of application of the laser beam is formed in a region which does not overlap with the positions of the internal wiring traces **232** as viewed from above. Accordingly, it is possible to avoid the possibility that, when the lead wiring portion **52**B is cut by using a laser beam or the like, the laser beam reaches the position of the internal wiring traces **232**, whereby the internal wiring traces **232** are cut by the laser beam.
(87) Notably, as in the first embodiment, the wiring substrate **201** has the the internal wiring traces **32** provided between the base ceramic layer **20** and the first ceramic layer **21**. The internal wiring traces **32** are disposed in such a manner that, as viewed from above, the internal wiring traces **32** detour around the regions where the alumina coats **55**A and **55**B are formed so as not to overlap with the regions where the alumina coats **55**A and **55**B are formed. As a result, it is possible to avoid the possibility that, when the lead wiring portions **52**A and **52**B are cut by using a laser beam or the like, the laser beam reaches the position of the internal wiring traces **32**, whereby the internal wiring traces **32** are cut by the laser beam.
(88) It should be considered that the embodiments disclosed this time are illustrative and are not restrictive in all aspects. It is intended that the scope of the present invention is shown by the claims rather than the above-described explanation and that the present invention encompasses all modifications within the meaning and scope equivalent to the claims. Also, the present invention encompasses configurations obtained by mutually combining the configurations of different embodiments described in the present specification.
REFERENCE SIGNS LIST
(89) **1**: wiring substrate **11**: ceramic substrate (insulating substrate) **20**: base ceramic layer **21**: first ceramic layer **22**: second ceramic layer **32**: internal wiring trace (wiring layer) **41**: solid pattern (internal metal layer) **42**: pattern removed region (opening) **52**: lead wiring portion (surface metal layer) **53**: terminal portion **55**: alumina coat (insulating coat layer) **61**: groove **70**: semiconductor package **71**: semiconductor element **101**: wiring substrate **111**: ceramic substrate **201**: wiring substrate **232**: internal wiring trace (wiring layer) **241**: solid pattern (internal metal layer)

Claims

1. A wiring substrate comprising: an insulating substrate; a surface metal layer provided on a surface of the insulating substrate and divided by a groove formed on the insulating substrate; and a wiring layer disposed in an inner part of the insulating substrate, wherein an insulating coat layer is provided around the groove, wherein the wiring layer is disposed in such a manner that, as viewed from above, the wiring layer detours around a region where the groove is formed and a region where the insulating coat layer is formed so as not to overlap with the region where the groove is formed and the region where the insulating coat layer is formed, and wherein an internal metal layer is further provided in an inner part of the insulating substrate, and the internal metal layer has an opening formed in a region which overlaps with the region where the groove is formed as viewed from above in such a manner that the opening contains the entirety of the region where the groove is formed.
2. The wiring substrate according to claim 1, wherein the internal metal layer is provided at a position located 0.5 mm or less from the surface of the insulating substrate.
3. A wiring substrate comprising: an insulating substrate; a surface metal layer provided on a surface of the insulating substrate and divided by a groove formed on the insulating substrate; and a wiring layer disposed in an inner part of the insulating substrate, wherein an insulating coat layer is

provided around the groove, wherein the wiring layer is disposed in such a manner that, as viewed from above, the wiring layer detours around a region where the groove is formed and a region where the insulating coat layer is formed so as not to overlap with the region where the groove is formed and the region where the insulating coat layer is formed, and wherein the wiring layer is provided at a position located 0.5 mm or less from the surface of the insulating substrate.